Dominik Szewczyk

Data Scientist · Machine Learning & AI

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PROFILE

Data Science and AI graduate (cum laude) who takes projects from start to finish. Across two internships and client work, I cover the whole process, from data analysis and machine learning to a working tool that meets the client's needs.

// 01 Experience

Data Scientist, Graduation Internship · KapiteinLabs

Dec 2025 - Jun 2026

- Built a predict-and-optimise machine learning system, delivered as a **no-code Streamlit dashboard** that engineers and clients use without writing code.
- Trained a physics-informed neural network (PINN) to predict ozone output from reactor settings, reaching **R^2 = 0.988, MAE = 1.49 g/kWh, MAPE = 1.9%** on 438 measurements from TU Eindhoven.
- Added an inverse optimiser that recommends the settings needed to hit a target output, replacing slow one-off lab experiments with instant estimates.

Data Scientist, Internship · De Rooy Transport & Logistics

Jan - Jun 2025

- Built a **multi-model pipeline** that forecasts where, when, and how large unexpected pickups will be (over 82% of pickups are unplanned same-day requests), now **in production** and feeding into route planning.
- Over a 5 km grid, a Keras classifier flags pickup cells (**90.2% accuracy**, 86.5% F1) and a random forest locates them (**18 m mean error**, 100% within 1 km), with further models for load size and insertion time.
- Engineered spatial, calendar, holiday, and lagged features. Route simulations showed up to **~1,500 km and ~900 kg CO_2 saved per day**; featured in a logistics-industry AI whitepaper.

Web Developer · Freelance

Feb 2026 - Present

Built and deployed 10 live websites for small-business clients, owning each project from design through to deployment.

// 02 Selected Projects

AI 3D Digital Twin & Object Removal

Antea Group & Kavel 10 · 2024 to 2025

Reconstructed urban environments from aerial and drone imagery. Benchmarked DeepFill v2, LaMa, and PowerPaint for object removal and U-Net, DeepLabV3, and FCN for segmentation, then built 3D meshes with Instant NeRF and a VR walkthrough in Unreal Engine.

Computer Vision · NeRF · Semantic Segmentation · 3D Reconstruction

Plant Analysis with Computer Vision & Robotics

NPEC · 2023 to 2024

Segmented seeds, shoots, and roots from plant images with a U-Net and instance segmentation, localising root tips to within roughly 1 mm. Simulated a PID-controlled liquid-handling robot and applied reinforcement learning to dispense precisely at the root tips.

Computer Vision · U-Net · Reinforcement Learning

ATHENA: MLOps for Plant Science

NPEC · 2024

Took ATHENA research models to production as a Python package and cloud app for organ segmentation and landmark detection. Built Azure pipelines, containerised with Docker, and served via FastAPI; recognised publicly by NPEC.

Microsoft Azure · Docker · FastAPI · MLOps

// 03 Education

FH Technikum Wien · Data Science & Artificial Intelligence (MSc)

2026 - Present

Part-time evening programme, available for full-time work during the day.

Breda University of Applied Sciences · Data Science & Artificial Intelligence

2022 - 2026

Graduated cum laude. Two industry internships and applied research in ML, computer vision, NLP, and MLOps.

II LO im. Andrzeja Frycza Modrzewskiego · Extended Mathematics

2019 - 2023

One of the top-ranked secondary schools in the region.

// 04 Skills & Tools

MACHINE LEARNING & AI	Machine Learning, Deep Learning, Computer Vision, NLP & Transformers, Predictive Modeling
LANGUAGES & TOOLS	Python, scikit-learn, TensorFlow, Keras, XGBoost, FastAPI, Streamlit, Power BI, R, Git
CLOUD & MLOPS	Microsoft Azure, Azure ML, Docker, CI/CD pipelines
WAYS OF WORKING	Agile (Scrum), data visualisation, stakeholder communication
LANGUAGES	English (fluent) · Polish (native) · German (basic)
INTERESTS	Football, gym, chess (3rd category), half marathon (1:43)